

Suggested Teaching Guidelines for
Web Based Java Programming
PGCP-AC, August 2026

Duration: 150 Hours: 60 Theory (T) + 60 Lab (L) + 30 Self Learning (SL)

Objective: To reinforce knowledge of Web technologies using Java.

Prerequisites: Strong knowledge of OOP with Java.

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment – 20%

List of Course Material

Text Book:

No specific courseware for the module, faculty may share some course materials.

Reference book:

1. Spring Boot in Action, 2015 edition, Craig Walls / O'Reilly
2. Java Persistence with Hibernate, Second Edition, Christian Bauer, Gavin King, and Gary Gregory / O'Reilly

Session 1: Introduction to JavaEE (2T + 2L + 2SL)

- JavaEE Overview
- JavaEE Container
- Packaging Web applications
- Web Servers and Application Servers
- JavaEE compliant web application
- Deployment tools.
- Web application life cycle

Sessions 2, 3, 4 & 5: Servlets and Session Management (8T + 8L + 4SL)

- Jakarta EE Overview
- Servlets: Dynamic Content Generation
- Servlet Life cycle
- Servlet API & Deployment
- Servlet Annotations
- The Servlet interface
- The HttpServlet, HttpServletRequest, HttpServletResponse
- Exception Handling
- Servlet, DAO, POJO DB Layers
- Session Management
- Session Tracking with
 - Cookies
 - HttpSession
- Request Dispatcher

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- Page Navigation
- Using Listeners
- Filters

Lab:

- Installing a servlet container (Tomcat)
- Adding Server to IDE
- Develop a structured dynamic web application (e.g. Library Management System) using servlets, deploy it in Tomcat
- Use HTTP Session in the Air Ticket Reservation System
- Compare which way of session tracking is better Cookies or HttpSession

Sessions 6 & 7: Introduction to MVC and JSP (4T + 4L + 2SL)

- MVC Architecture
- JSP: Separating UI from Content generation code
- Life cycle of a JSP page
- Directives, Implicit and Explicit Objects, Scriptlets, Expressions, Expression Language
- Scope
- JSP Error Page handling
- JSTL
- Creating a custom tag library

Lab:

- Separate UI code from the controller code in your Library Management System by incorporating JSP and Servlets.
- Complete the implementation of Air Ticket Reservation System.
- Implement MVC based web application using Servlet, JSP

Sessions 8, 9, 10 & 11: Maven and Hibernate Framework (8T + 8L + 4SL)

- Basic Introduction of MAVEN
- Maven Build Life Cycle
- Hibernate Framework
 - Introduction to Hibernate Framework
 - Architecture
- Hibernate in IDE
 - Creating web application using Hibernate API
 - Lifecycle of Hibernate Entities
- HB with annotation example
- Hibernate Mappings and Relationships
- Collection and Component Mapping
- HQL, JPQL

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Lab:

- Demonstrate Hibernate as standalone library in Java application
- Develop a web application (Online Bookshop) using Hibernate Persistence

Sessions 12, 13, 14 & 15: Spring Framework (8T + 8L + 4SL)

- Overview of Spring Architecture
- Spring MVC architecture
- Spring Modules Overview
- Understanding Spring 4 annotations (Basic Introduction)
- IoC (Inversion of Control)
- IOC container
- Dependency Injection
- Spring Beans
- Autowiring Beans
- Bean Scopes
- Spring MVC
- Model, Model & View, HandlerMapping, ViewResolver
- Design Pattern: Front Controller Pattern
- Spring MVC Web application with JSP views (without Spring Boot)
- Using Thymeleaf as alternate View Technology (only introduction)
- Spring Validations
- Introduction to Spring i18n and Localization

Lab:

- Design and deploy Library Management System using Spring Web

Sessions 16, 17, 18 & 19: Spring Boot (8T + 8L + 4SL)

- Spring Boot essentials
- Need of Spring boot
- Spring Boot Overview
- Building Spring Web application with Boot
- Spring Boot in detail (Use Spring Boot for all demo & assignments here onwards)
- Running a web application using Spring Boot with CRUD (with Static Data not DB)

Lab:

- Create Hello World Spring Boot Web application
- Check Libraries imported by Spring Boot
- Create Spring Boot CRUD application with View technology

Sessions 20 & 21: Spring Data module (4T + 4L + 2SL)

- Spring Data JPA (Repository support for JPA)
- CrudRepository & JpaRepository

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- Query methods
- Using custom query (@Query)

Lab:

- Add CRUD operations with Spring JPA etc. to earlier Spring Web application.

Session 22: Spring AOP (2T + 2L)

- AOP Overview
- Spring AOP
- AOP Terminology and annotations: Advice, Join Points, Pointcuts, Aspects

Lab:

- Modify earlier Spring MVC application to Log all the requests using AOP

Sessions 23 & 24: Building REST services with Spring (4T + 4L + 2SL)

- Introduction to web services
- RESTful web service introduction
- Create RESTful web service in java using Spring Boot
- RESTful web service JSON example
- RESTful web service CRUD example
- Using POSTMAN client to invoke REST API's
- Integrating Java Mail API from Spring Boot
- Open API/Swagger documentation

Lab:

- Create REST API for Employee Management using Spring Boot
- Invoke it from POSTMAN app
- Invoke it from another Spring Boot Web application using REST Template

Session 25: Testing in Spring (2T + 2L + 2SL)

- Testing in Spring
- Unit Testing of Spring Data Layer
- Unit Testing of Spring Service Layer
- Integration Testing of Spring REST API
- Unit Testing Spring MVC Controllers with REST

Sessions 26 & 27: Spring Security (4T + 4L + 2SL)

- What is Spring Security
- Spring Security with Spring Boot
- Basic Authentication

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- Authentication with User credentials from Database and Authorization
- JWT Authorization

Lab:

- Design & Test Spring Application
- Secure the Spring Web application created in earlier exercise.

Sessions 28, 29 & 30: Microservices (6T + 6L + 2SL)

- Monolithic vs Microservices
- Microservices Architecture
- Fragmentation of business requirement
- Database Management for Microservices
- Synchronous Communication between microservices using RestTemplate and OpenFeign
- Deployment pattern
- Service Registry and Discovery
- API gateway

Lab:

- Demonstration of Spring Boot Microservices